BIF604 – Special Topics in Bioinformatics

Midterm 2025

Objective + Subjective File

OBJECTIVES SECTION

- **Gene omnibus (GO) is for. Select the best option**

A. Genome

B. ✔Gene and Transcriptiomics

C. Transcriptiomics

D. Protein

- **Linux is a... Select the best option**

A. ✔Operating system

B. Compiler

C. All

D. Platform

- **Sequence Alignment Map format A TAB-delimited text format consisting of a header section and Alignment body/section**

Answer: SAM

- **GTF sequences are not included. Select the best option**

A. ✔ <rate>

B. <end>

C. <start>

D. <score>

- **A For loop is a repetition control structure that allows you to efficiently write a loop that needs to execute a specific number of times. Like a _____ statement, except that it tests the condition at the end of the loop body.**

Answer: while

- **GEO has _____ kinds of data records**

Answer: Four

- **_____ Non-Sanger non fluorescence technique that quantitatively measures released PPi**

Answer: Pyrosequencing

- **The slide is occupied by a large number of nucleotides and DNA polymerases**

Answer: SBS (Sequencing By Synthesis)

- **Removing individual sequences**

Answer: NGS: QC and manipulation → Filter FastQ [under Generic FASTQ manipulation]

- **<seqname> <source> <feature> <start> <end> <score> <strand> <frame> [attributes] [comments]**

Answer: GTF format

- **Strength of R are except. Select the best option:**

A. Compact expression of ideas

B. Operation on model objects,

C. ✔Semantic complications

D. Seamless integration of new code

- **Non-Sanger nonfluorescence technique that quantitatively measures released PPi**

Answer: Pyrosequencing

- **SAM (Sequence Alignment Map) format is a _____ based format that stores alignment data.**

Answer: text

- **_____ a single file using wget**

Answer: Download

- **BAM (Binary Alignment Map) format is the binary version of_____**

Answer: SAM

- **Advantage of the galaxy except:**

Answer: Installation of new software and writing new tools usually requires use of the command line

- **Disadvantage of the galaxy except. Select the best option:**

A. User-friendly interface with drop-down menus

B. Workflows facilitate processing data

C. All Galaxy tools are available for the command line

D. ✔Both A & B

- **The input sample is first cleaved into short length _____ fragments.**

Answer: 100-150 bp

- **Biology + chemistry + physics + computer science + data science + computer programming + Mathematics and Statistics is_____**

Answer: Bioinformatics

- **BWA gives output in_____**

Answer: SAM file

- **Maps available in Galaxy. Select the best option**

A. BWA

B. Bowtie

C. Whisper 2

D. ✔All of them

- **In the history panel of Galaxy we can't do**

A. Edit

B. ✔Colour

C. Delete

D. View

- **_____ uses a directed acyclic graph algorithm to identify the minimum set of independent transcripts that can explain the reads observed in an RNA-Seq experiment.**

Answer: CuffLinks

- **FastaQ quality control.**

Answer: All of these

- **Strength of R are except: _____, Compact expression of ideas, Operation on model objects, Seamless integration of new code**

Semantic complications

- **Packages for differential gene expression analysis:**

Answer: All of these

- **BAM (Binary Alignment Map) format is the binary version of _____**

SAM

- **The input sample is first cleaved into short length fragments**

100-150 bp

SUBJECTIVE SECTION

Question 1: What are the packages of R language used in bioinformatics?

These R packages are used to process, analyze, or visualize genomic and transcriptomic data, especially in gene expression studies:

1. Bioconductor
2. Limma
3. EdgeR
4. DESeq
5. GEOquery
6. Genomic Features
7. org.Hs.eg.db

Question 2: What are NGS data formats?

Common formats for storing sequencing data:

- **FASTA:** Stores nucleotide sequences.
- **FASTQ:** Stores sequences along with quality scores.
- **SAM:** Text format for storing aligned reads.
- **BAM:** Compressed binary version of SAM.

Question 3: What is the significance of quality control in mapping?

- Filters out low-quality reads and adapter sequences
- Ensures clean input for mapping to avoid false alignments
- Leads to accurate and reliable downstream analysis

Question 4: What formats are supported by Galaxy?

Galaxy tools accept various standard NGS file formats including:

- FASTQ (raw reads)
- FASTA (reference sequences)
- SAM/BAM (aligned reads)
- VCF (variant calls)
- GTF (gene annotation)

Question 5: What are the types of loops in R programming?

1. **For loop –** Repeats a task for a specific number of times.
2. **While loop –** Repeats as long as a condition is true.
3. **Repeat loop –** Repeats indefinitely until a break command is given.

Question 6: What is the role of a phylogenetic tree?

- It is a diagram that shows evolutionary relationships between species or genes
- Identifies common ancestors
- Helps study mutations and gene families.
- Useful for disease tracing and evolutionary biology

Question 7: What is Biopython and what is it used for?

Biopython is a Python-based toolkit for bioinformatics that allows users to:

- Handle DNA, RNA, and protein sequences.
- Read/write formats like FASTA and GenBank.
- Perform sequence alignments.
- Access biological databases like NCBI and BLAST

Question 8: What is microarray and what are its types?

A microarray is a laboratory tool to measure gene expression by hybridizing mRNA to DNA probes on a chip. It works by fixing thousands of DNA probes on a chip and then observing how much mRNA from a sample bind to each other.

Types include:

1. cDNA Microarray
2. Oligonucleotide Microarray
3. SNP Microarray
4. Protein Microarray

Question 9: What are the basic data types in R?

1. **Numeric** – Decimal numbers
2. **Integer** – Whole numbers
3. **Complex** – Numbers with imaginary part
4. **Logical** – TRUE or FALSE
5. **Character** – Text values

Question 10: What do you know about bioinformatics?

Bioinformatics is the use of computer tools and software to analyze biological data like DNA, RNA, and protein sequences. It combines biology, statistics, and programming to study gene functions, mutations, and molecular structures.

- Use **tools like** Galaxy, R, Python, and Linux to study genes and their functions
- **Perform tasks** like gene expression analysis, sequencing, and mutation detection
- Work with **databases like** GEO and file formats like FASTA, SAM, BAM
- It **helps scientists** understand how genes behave, how diseases spread, and how to develop treatments using genetic data.

Question 11: What is the difference between SAM and BAM?

Both SAM and BAM are file formats used in bioinformatics to store aligned sequencing reads (after mapping to the reference genome), but:

SAM (Sequence Alignment Map)	**BAM (Binary Alignment Map)**
SAM is a text format	BAM is a compressed binary version of SAM
Human-readable	Not human-readable
Easy to open and view in Notepad or any text editor	Requires special tools to view (like samtools or IGV)
Larger in size	Faster to load and smaller in size

Question 12: How is mapping done in Linux in bioinformatics?

Mapping in LINUX is the process of matching short DNA reads to a reference genome using command like tools. This is done to find out where each read belongs in the original genome.

The process includes the following steps:

1. Index the reference genome using BWA index
2. Align reads using bwa mem and generate SAM file
3. Convert to BAM using samtools view
4. Sort and visualize with tools like IGV

Question 13: What are the limitations of microarray tool?

- Detects only known genes (no novel discovery)
- Cross-hybridization can cause errors.
- Low sensitivity for low-expressed genes
- Requires large sample amount
- Limited dynamic range
- Replaced in many areas by RNASeq

Question 14: What is Sequencing by Synthesis (SBS)?

A method used in Illumina technology where:

- DNA is fixed and a polymerase adds bases one by one.
- Each base emits a signal when added.
- Signals are detected cycle by cycle to form the full sequence.

Question 15: What are three possible input formats in Galaxy?

Common input formats in Galaxy are:

1. **FASTQ** – for raw sequencing reads
2. **FASTA** – for reference sequences
3. **GTF** – for gene annotations

Question 16: Pros and Cons of Galaxy with attributes

Attribute	Pros	Cons
User Interface	Easy, graphical	Limited for advanced users
Tool Integration	Many pre-installed tools	Can crash with large files
Reproducibility	Workflows can be saved	Workflow sharing limited
Data Management	Supports multiple file types	Storage is limited
Accessibility	Web-based, no install	Requires internet
Visualization	Simple built-in viewers	Limited compared to IGV

Question 17: How does BWA do mapping in Linux?

Steps of Mapping in LINUX using BWA:

1. Index the reference genome (bwa index)
2. Align reads (bwa mem)
3. Output in SAM format
4. Convert and sort as BAM using samtools

Question 18: What does GO stand for and describe it?

GO = Gene Ontology

Gene Ontology is a standardized system to describe gene function using 3 categories:

1. **Biological Process:** What the gene product does
2. **Molecular Function:** The actual activity of the gene product
3. **Cellular Component:** Where in the cell the gene product works

Question 19: How to make phylogenetic tree of multiple sequence?

A phylogenetic tree is a diagram that shows the evolutionary relationship between different organisms or sequences. It is made after multiple sequence alignment (MSA) to understand how similar or different the sequences are.

Steps to Make It:

- **Step 1:** Perform multiple sequence alignment of DNA/protein sequences
- **Step 2:** Use tools like Clustal Omega or MEGA to compare
- **Step 3:** Generate a phylogenetic tree based on sequence similarity

Question 20: Describe Python and its uses?

Python is a free and open-source programming language used in bioinformatics for tasks like:

- Writing analysis scripts
- Handling FASTA/FASTQ files
- Accessing databases (like NCBI)
- Building tools for gene analysis

It is popular because it is easy to learn, has many libraries (like Biopython), and works well with biological data.

Question 21: Describe 4 types of GEO

GEO (Gene Expression Omnibus) is an NCBI database that stores gene expression and functional genomics data. In GEO, we generally deal with four common types of accession IDs that are:

1. **GSE (Series):** A group of related samples or experiments
2. **GSM (Sample):** A single biological sample used in an experiment
3. **GPL (Platform):** The technology or microarray chip used for measuring expression
4. **GDS (DataSet):** Curated datasets derived from series

Question 22: Describe String and also mention its main features

In programming (including R), a string is a sequence of characters. It is used to store text like gene names, file names, or any alphabetic data.

Main Features of String:

1. **Text-based Data:** Stores words, names, or letters
2. **Immutable:** Once created, strings cannot be changed
3. **Indexed:** Characters in a string can be accessed by position
4. **Operations Supported:** You can perform operations like joining, slicing, replacing, and searching inside strings
5. **Quotes Needed:** Strings are always written inside single or double quotes: "hello" or 'hello'

Question 23: What are arrays in programming? Explain their applications in bioinformatics.

In programming, an array is a data structure used to store multiple values of the same data type in a single variable. It can be 1D (vector), 2D (matrix), 3D or more (like layers of matrices. Arrays allow efficient data storage, retrieval, and processing, especially when dealing with large amounts of similar data.

Applications of Array:

- To Store Gene Expression Data
- To Work with DNA or Protein Sequences
- For Image Storage
- To Do Calculations
- To Repeat Tasks Easily
- To Simulate or Model Systems

Question 24: Write the Principle of Comparative Genomic Sequencing

Comparative Genomic Sequencing is the process of comparing the genome sequences of different organisms or individuals to find similarities and differences.

Principles of Comparative Genomic Sequencing:

- By comparing genomes, we can find DNA regions that are shared (conserved) between species

- Helps in finding mutations, SNPs, insertions, deletions etc., which can be related to diseases or traits
- Genomic comparison shows how closely related two organisms are — used in phylogenetics.
- If a gene in one species is similar to a known gene in another, its function can be predicted.
- Comparing with well-annotated genomes helps annotate new genomes more accurately.

Question 25: What are three steps in analyzing gene expression data in microarray?

Gene expression analysis using microarray is done in three main steps:

1. **Sample Preparation and Hybridization:** mRNA is collected from cells and converted into cDNA. Then, cDNA is labelled with fluorescent dye. This cDNA is applied to a microarray chip containing DNA probes.
2. **Scanning and Data Collection:** After hybridization, the chip is scanned using a laser. The light (fluorescence) emitted shows which genes are active. Higher brightness = higher gene expression.
3. **Data Analysis and Interpretation:** Signal intensities are converted into expression values**.** These values are compared across samples (e.g., normal vs. diseased). The result shows which genes are up-regulated or down-regulated

Question 26: What Tools used in DAVID?

DAVID stands for Database for Annotation, Visualization, and Integrated Discovery. It is a free online bioinformatics resource used to analyze large gene or protein lists.

Main Tools/Features in DAVID:

1. Functional Annotation Tool
2. Gene Functional Classification Tool
3. Functional Annotation Clustering
4. Chart Report
5. Pathway Analysis
6. Gene ID Conversion Tool

Question 27: Explain Genome Sequencing?

Genome sequencing is the process of determining the complete DNA sequence of an organism's genome. It tells us the order of nucleotides (A, T, G, C) in all chromosomes.

Main points about Genome Sequencing:

1. It reveals the entire genetic code of an organism
2. Helps in identifying mutations, genes, and regulatory elements
3. Used in medicine, agriculture, and research
4. Modern techniques like Next Generation Sequencing (NGS) make it faster and cheaper
5. Results are saved in FASTA, FASTQ, or BAM formats

Question 28: Write the Attributes of "History Panel" of Galaxy

The History Panel in Galaxy is the right-hand side panel that records and displays all the datasets and analysis steps performed during a session.

Main Attributes History Panel:

1. **Tracks Analysis Steps:** Lists all datasets uploaded or generated during analysis. Displays them in the order they were created, with the newest at the top.
2. **Dataset Management:** Allows users to view, rename, delete, or annotate datasets. Provides options to manage dataset visibility and status
3. **Multiple Histories:** Users can create and manage multiple histories for different projects.
4. **Reproducibility:** Maintains a complete record of all analysis steps, ensuring reproducibility of results
5. **Export and Sharing:** Histories can be exported, shared with other users, or used to extract workflows.

Question 29: What is a key advantage of Galaxy?

Galaxy is web-based and user-friendly, allowing researchers to run complex bioinformatics tools without coding. It supports workflow creation, reproducibility, and easy data management.

Question 30: What is a disadvantage of Galaxy?

Galaxy can be slow with large datasets, has limited customization, and often requires a stable internet connection. Public Galaxy servers may have storage limits or crash under heavy use.

Question 31: Write the Types of formats of R?

In R, the term “formats” usually refers to data types and data structures used to store and process values.

Basic Data Types in R:

- Numeric
- Integer
- Complex
- Logical
- Character

Question 32: Describe sequence alignment?

Sequence alignment is the process of arranging two or more DNA, RNA, or protein sequences to identify regions of similarity or difference.

This is useful for finding mutations, Understanding evolutionary relationships, Comparing gene/protein function

Types of Alignment:

1. Pairwise Alignment
2. Multiple Sequence Alignment (MSA)
3. Global Alignment
4. Local Alignment

Question 33: What is multiple sequence alignment (MSA)?

Multiple sequence alignment is the process of aligning three or more DNA, RNA, or protein sequences to find regions that are similar or conserved.

It helps in:

- Detecting functionally important regions
- Building phylogenetic trees
- Identifying mutations or evolutionary changes

Question 34: Define phylogenetic and evolutionary tree?

A phylogenetic tree (also called an evolutionary tree) is a diagram that shows how different organisms or sequences are related to each other based on their DNA, RNA, or protein similarities.

Question 35: Write about the File in Python

In Python, a file is used to store data permanently, such as text, FASTA sequences, or analysis results. Python allows us to Open a file, Read from it, Write to it, Close it after use

Question 36: Describe the "Dictionary" in Python

- A dictionary in Python is a data structure that stores data in key-value pairs.
- It is used when you want to associate a value with a name or label (key).
- Used in bioinformatics for things like gene names, expression values, metadata

Question 37: How multiple sequence alignment affects evolutionary tree?

Multiple sequence alignment plays an important role in building accurate evolutionary (phylogenetic) trees.

How MSA Affects the Tree:

- MSA finds similar regions across sequences, which help identify common ancestors.
- Helps in locating insertions, deletions, and substitutions, which show how species evolved differently.
- Better alignment = more reliable phylogenetic tree
- The more similar two sequences are in MSA, the closer they appear in the tree.

Question 38: How can data be graphically represented and visualized in R language? Explain with types of graphs.

In R programming, data visualization is an essential part of data analysis. Graphical representation helps to understand patterns, trends, distributions, and relationships in datasets. R provides several built-in functions and packages (like ggplot2) for creating a variety of graphs.

Types of Graphs in R:

1. Bar Plot

- Used to compare values across different categories
- Function: barplot ()

2. Line Plot

- Used to display trends over time
- Function: plot(x, y, type = "l")

3. Histogram

- Displays the distribution of continuous numerical data
- Function: hist()

4. Boxplot

- Used to show spread, central tendency, and outliers in the data
- Function: boxplot()

5. Pie Chart

- Shows the proportion of categories in a dataset
- Function: pie()

Question 39: Write a short note on Differential Gene Expression

Differential Gene Expression (DGE) is a fundamental concept in bioinformatics and molecular biology. It involves analyzing how gene expression levels change between two or more biological conditions.

It compares gene activity across samples, such as normal vs. diseased tissues. It helps in understanding disease mechanisms, biomarker discovery, and drug response.

Question 40: What are the Applications and Advantages of Pyrosequencing?

Pyrosequencing is a sequencing technique based on the detection of released pyrophosphate during DNA synthesis. It is known for its speed, real-time detection, and accuracy in short-read sequencing.

Applications of Pyrosequencing:

- Mutation and SNP detection in clinical diagnostics
- DNA methylation analysis (epigenetic studies)
- Microbial identification in infection diagnosis

Advantages of Pyrosequencing:

- Provides real-time sequence data
- High accuracy for short sequences
- No need for gel electrophoresis